SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

COMPUTER SCIENCE DEPARTMENT

END OF SEMESTER EXAMINATIONS

COURSE: MULTIMEDIA SYSTEMS
COURSE CODE: CIT 4541

TIME ALLOWED: 3 Hours
TOTAL MARKS: 100 Marks

GENERAL INSTRUCTIONS TO CANDIDATES:

1. This paper contains **FIVE (5)** questions, each carrying **25 marks.**
2. Answer **ANY FOUR (4)** questions.
3. Each question has sub-questions labelled (a) to (e), each carrying **5 marks.**
4. **Neat and clear work** is highly recommended.
5. Write your student number clearly on all answer sheets.
6. Begin each question on a new page.

QUESTION 1

(a) Define the term **Multimedia** and identify FOUR (4) elements that make up a multimedia presentation. For each element, state ONE (1) example of software used to create or handle it. **[8 marks]**

(b) Study the diagram below showing the **Global Structure of a Multimedia System** and answer the questions that follow.

APPLICATION DOMAIN (Tools, Applications, Documents, User Interface, Programming Abstraction)
SYSTEM DOMAIN (Operating Systems | Communication Systems | Database Systems)
DEVICE DOMAIN (Sound | Image/Graphics | Video | Animation | Storage | Network | Compression)

Figure 1.1: Global Structure of a Multimedia System

(i) Briefly describe the role of each of the three domains shown in Figure 1.1. **[6 marks]**

(ii) State the domain in which **synchronisation** is most critical and justify your answer. **[3 marks]**

(c) Distinguish between **interactive multimedia** and **non-interactive multimedia**, giving a practical example of each. **[4 marks]**

(d) Multimedia is described as an *inter-disciplinary subject*. Identify and briefly explain THREE (3) disciplines involved. **[4 marks]**

QUESTION 2

(a) With the aid of a clearly labelled diagram, describe the **digitisation process** of an analogue sound wave and explain the role of the **Nyquist Sampling Theorem** in determining sampling rate. **[7 marks]**

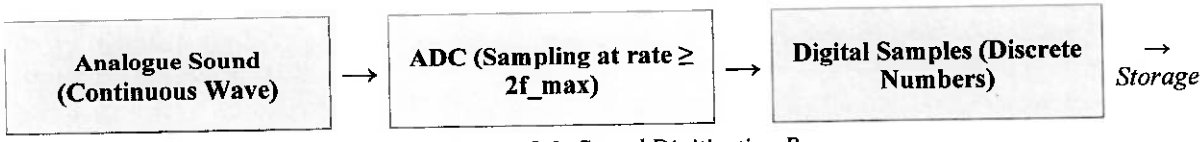


Figure 2.1: Sound Digitisation Process

(b) Using the formula $\mathbf{S = R \times (b/8) \times C \times D}$, calculate the file size of the following audio recordings. Show all steps and express your final answers in **Megabytes (MB)**. **[8 marks]**

Recording	Sampling Rate (R)	Resolution (b)	Channels (C)	Duration (D)
Recording 1	44,100 Hz	16 bits	2 (Stereo)	3 minutes
Recording 2	22,050 Hz	8 bits	1 (Mono)	5 minutes
Recording 3	48,000 Hz	24 bits	2 (Stereo)	1 minute 30 sec

Table 2.1: Audio Recording Parameters

Note: 1 KB = 1024 bytes; 1 MB = 1024 KB

(c) Compare **MIDI music** with **Digitised Audio** under the following headings: file size, editability, sound quality, and hardware requirements. **[6 marks]**

Criterion	MIDI Music	Digitised Audio
File Size		
Editability		
Sound Quality		
Hardware Requirements		

Table 2.2: MIDI vs. Digitised Audio (Complete the table)

(d) Define **dynamic range** and calculate the dynamic range (in dB) of an orchestra that reaches **130 dB** at fortissimo and drops to **30 dB** at pianissimo. Use the decibel formula: **$dB = 10 \log(I2/I1)$**. **[4 marks]**

QUESTION 3

(a) A digital image is described as **24-bit true colour** with a resolution of **800 × 600 pixels**. Using clear calculations:

(i) Determine the total number of pixels in the image. **[2 marks]**

(ii) Calculate the file size of the uncompressed image in **kilobytes (KB)**. **[3 marks]**

(iii) If displayed on a monitor with **96 dpi** resolution, calculate the **physical dimensions** (width × height in inches) of the displayed image. **[3 marks]**

(b) Distinguish between **bitmap (raster) images** and **vector graphics**. Use the table below to structure your answer. **[6 marks]**

Criterion	Bitmap Image	Vector Graphic
Data Representation		
Scalability		
File Size		
Best Use Cases		
Common File Formats		

Table 3.1: Bitmap vs. Vector Graphics

(c) Study Figure 3.1 below showing the **RGB Colour Model** and answer the questions that follow.

Colour Name	RED (0–255)	GREEN (0–255)	BLUE (0–255)	Resulting Colour
Red	255	0	0	
Green	0	255	0	
Blue	0	0	255	
Yellow	255	255	0	
Cyan	0	255	255	
Magenta	255	0	255	
White	255	255	255	
Black	0	0	0	
Light Gray	127	127	127	

Figure 3.1: RGB Colour Values (fill in the Resulting Colour column)

(i) Complete the **Resulting Colour** column in Figure 3.1. **[3 marks]**

(ii) How many total colours can a **24-bit** true colour image represent? Show your calculation. **[2 marks]**

(d) State **FOUR (4) ways** a digital image can be acquired and give ONE real-world application for each. **[4 marks]**

(e) Explain the difference between **image resolution, display resolution,** and **output resolution**. Why is it important to match these when preparing images for print? **[2 marks]**

QUESTION 4

(a) Explain the difference between a **typeface** and a **font**. Give ONE example of each and state the **FIVE (5) attributes** used to specify a font. **[6 marks]**

(b) Study Figure 4.1 below which illustrates key measurements used in typography. Label the diagram using the terms provided in the word bank.

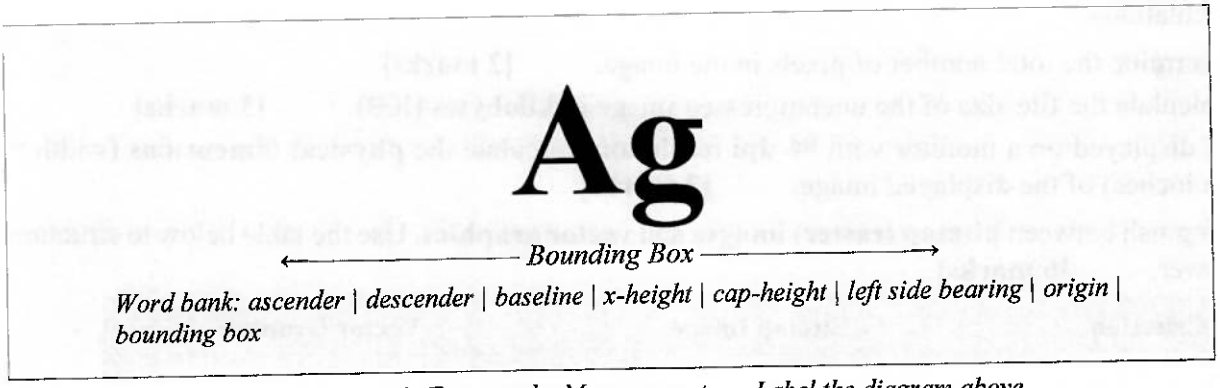


Figure 4.1: Typography Measurements — Label the diagram above

[5 marks]

(c) Explain the difference between **serif** and **sans serif** typefaces. For each type:

(i) Draw or describe a distinguishing visual feature. **[3 marks]**

(ii) State the recommended use in multimedia vs. print. **[2 marks]**

(d) Define the following text layout terms and state the units used for measurement:

Term	Definition	Unit of Measurement
Leading		
Tracking		
Kerning		

Table 4.1: Text Layout Terms

[4 marks]

(e) Compare **bitmap fonts** and **outline fonts**. Explain why outline fonts are preferred in modern multimedia systems. **[5 marks]**

QUESTION 5

(a) Explain the following visual perception phenomena that make motion video possible. Use a clearly labelled diagram to support your explanation of at least ONE of them. **[6 marks]**

(i) Persistence of Vision

(ii) Phi Phenomenon

(iii) Temporal Aspect of Illumination (Fusion Frequency)

(b) Study Table 5.1 below showing specifications for three broadcast television systems and answer the questions that follow.

System	Total Scan Lines	Active Resolution	Vertical Resolution	Horizontal Resolution	Total Pixels
NTSC	525	484	242	330	106,000
PAL	625	575	290	425	165,000
SECAM	625	575	290	465	180,000

Table 5.1: Broadcast Television Systems Comparison

(i) Which system has the highest total pixel count and what is the significance of this? **[2 marks]**

(ii) Using the PAL system (**352 × 288 resolution, 16 bits per pixel, 25 frames/sec**), calculate the **raw video data rate** in both **Mbit/s** and **MB/s**. Show all working. **[5 marks]**

(iii) Given a standard Ethernet bandwidth of **10 Mbit/s** and a double-speed CD-ROM rate of **300 KB/s**, explain why video data **must be compressed** before storage or transmission. **[3 marks]**

(c) With the aid of a labelled **block diagram**, describe the components of a **Video Capture Card** and the function of each component. **[5 marks]**



Figure 5.1: Video Capture Card Block Diagram — Describe the function of each component

(d) In computer animation, distinguish between **forward kinematics** and **inverse kinematics (IK)**. Explain with an example how IK is used to animate a human arm. **[4 marks]**

— END OF EXAMINATION PAPER —

Good Luck to All Candidates